

Section 14.1: Functions of Several Variables

Partial Derivatives

MTH 310
Calculus III

Functions of Several Variables

Function of Two Variables

A **function of two variables** is a rule that assigns to each ordered pair (x, y) in a set $D \subseteq \mathbb{R}^2$ a unique real number z :

$$z = f(x, y)$$

- **Domain:** the set D of valid inputs (x, y)
- **Range:** the set of all output values $z = f(x, y)$

Same idea as Calc I, but now the input is a **point** in $\mathbb{R}^2$ (or $\mathbb{R}^3$) instead of a single number.

Example 1: Finding the Domain

Example 1

Find and sketch the domain of $f(x, y) = x \ln(x^2 - y)$.

Look for restrictions, set up and solve inequalities, sketch.

Your Turn: Find the Domain

Example 2

Find the domain of $g(x, y) = \frac{\sqrt{y - x^2}}{1 - x^2}$.

Identify both restrictions, then describe and sketch the domain.

Graphs of Functions of Two Variables

Graph of a Function of Two Variables

The **graph** of f is the set of all points (x, y, z) in $\mathbb{R}^3$ such that $z = f(x, y)$ and $(x, y) \in D$.

$$\text{graph} = \{(x, y, z) \mid z = f(x, y), (x, y) \in D\}$$

Input = point in the xy -plane. Output z = height above (or below) that point. The collection of all these heights forms a **surface**.

Example 3

Sketch the graph of $f(x, y) = -3x - 2y + 6$.

Find the intercepts and sketch the plane.

Traces: Slicing a Surface

Traces

A **trace** of a surface is the curve of intersection when the surface is sliced by a plane.

- **Horizontal trace** (or *cross-section*): set $z = c$ (a constant) and see what curve remains in the xy -plane
- **Vertical trace**: set $x = c$ or $y = c$ and see the resulting curve

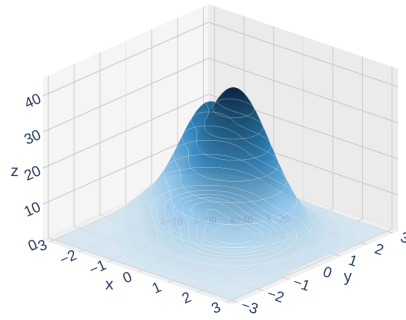
Traces reduce a 3D problem to a **2D curve** — something you already know how to analyze from Calc I.

Level Curves and Contour Maps

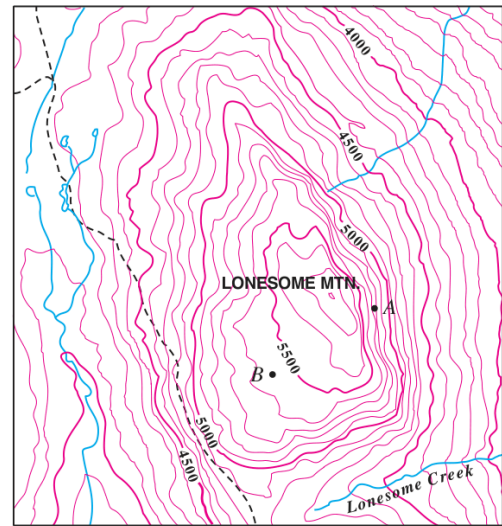
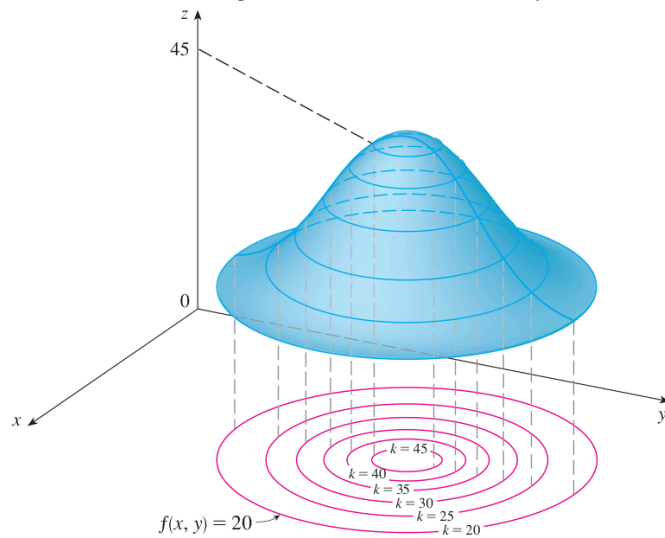
Level Curves

The **level curves** of $f(x, y)$ are the horizontal traces $f(x, y) = k$ for various constants k .

Each level curve is drawn in the xy -plane. Viewed together, they form a **contour map** (or *contour plot*).



Drag to rotate --- see how the surface produces the contour map on the floor | [Interactive version](#)



Contour lines show constant elevation

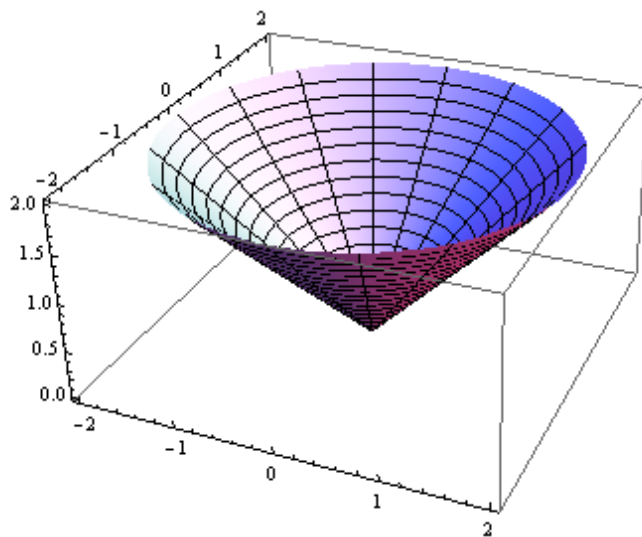
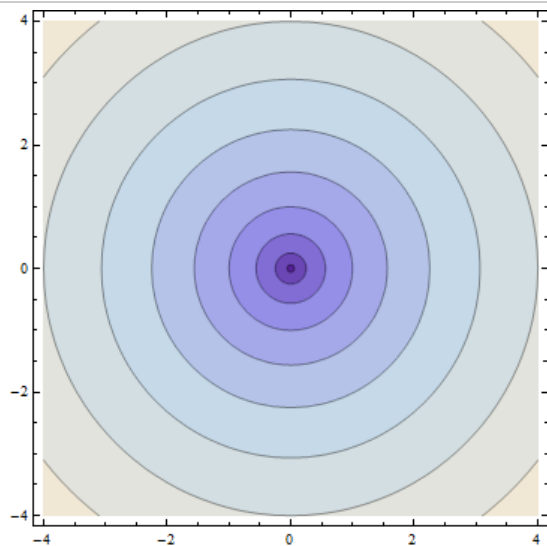
A contour map compresses 3D information into a 2D picture. Reading one is like reading a topographic map — each curve is a constant height $z = k$.

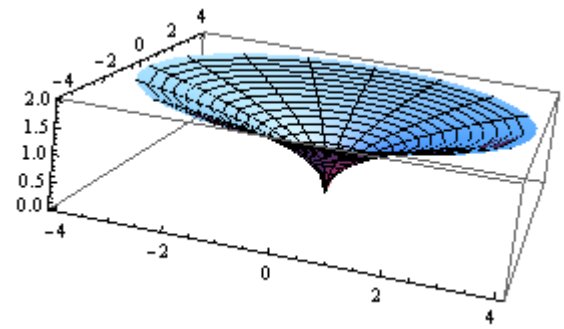
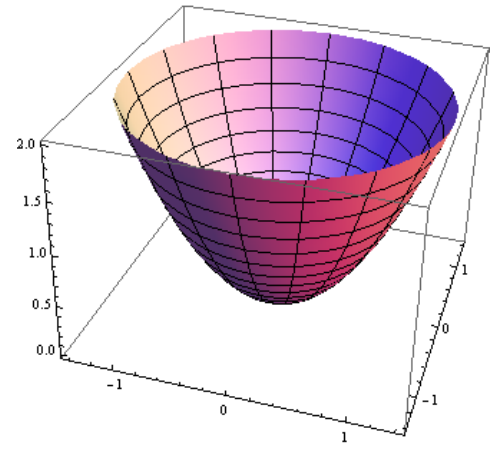
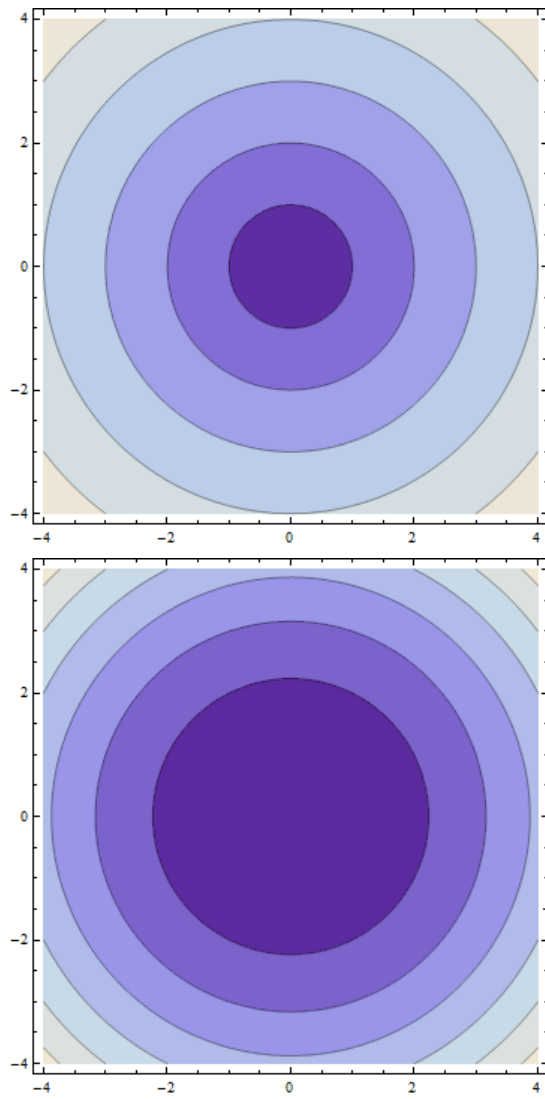
Example 4: Contour Plot of $f(x, y) = x^2 - y^2$

Example 4

Sketch the level curves of $f(x, y) = x^2 - y^2$ for $k = -4, -1, 0, 1, 4$.

Match the Contour Plot to the Surface

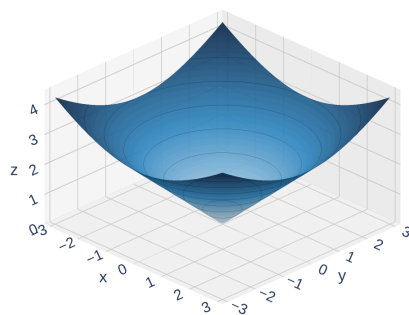




Write your matches and explain your reasoning.

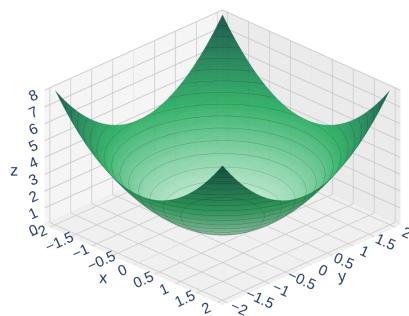
The Equations Behind the Surfaces

$$z = \sqrt{(x^2 + y^2)}$$



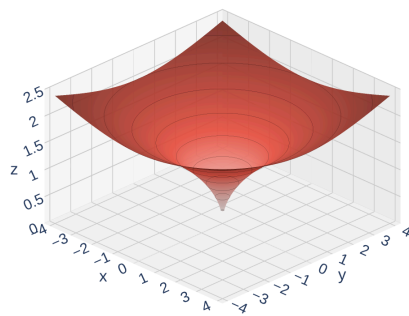
[Interactive version](#)

$$z = x^2 + y^2$$



[Interactive version](#)

$$z = (x^2 + y^2)^{1/4}$$



[Interactive version](#)

Level Surfaces: Functions of Three Variables

Level Surfaces

The **level surfaces** of $f(x, y, z)$ are the surfaces defined by $f(x, y, z) = k$ for constants k in the range of f .

Level curves collapse a 3D graph to 2D. Level surfaces collapse a 4D graph to 3D. Both let us visualize functions we can't directly graph.

Example 6: Level Surfaces of a Temperature Field

Example 6

A temperature distribution is given by:

$$T(x, y, z) = \frac{100}{(x^2 + y^2 + z^2)^{3/2}}$$

Describe the level surfaces. What shape are the level surfaces? What happens to temperature as you move away from the origin?

Set $T = k$ and solve for the relationship between x , y , and z .

Homework

Section 14.1: 3, 9, 13, 23-31, 32, 36, 41, 42, 45, 47, 49, 61-66, 67, 68